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Physical Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Liquid Solution

English Medium

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LIQUID SOLUTION

1. SOLUTION

It is the homogeneous mixture of two or more components.

The substance which dissolve other substance is **solvent** & the substance which is dissolved is **solute**, independent of their quantity. If both are soluble in each other then the substance present in larger amount by mole is solvent.

* A solution may exist in any physical state.

Types of Solution :

	Solvent	Solute	Examples
1.	Gas	Gas	Mixture of gases, eg. air.
2.	Gas	Liquid	$CHCl_3(\ell) + N_2(g)$
3.	Gas	Solid	Camphor (s) + $N_2(g)$.
4.	Liquid	Gas	CO_2 gas dissolve in water (aerated drink), soda water.
5.	Liquid	Liquid	Mixture of miscible liquids e.g. alcohol in water.
6.	Liquid	Solid	Salt in water, sugar in water.
7.	Solid	Gas	hydrogen over palladium.
8.	Solid	Liquid	Mercury in zinc, mercury in gold, i.e. all amalgams.
9.	Solid	Solid	Alloys e.g. copper in gold, zinc in copper.

2. SOLUBILITY

Maximum amount of solute which can be dissolved in a specified amount of solvent at constant temperature is its solubility. Solubility is affected by

1. nature of solute and solvent
2. temperature and
3. pressure

3. SOLUBILITY OF SOLID IN A LIQUID

Polar solutes are soluble in polar solvent and non polar solutes are soluble in non polar solvent due to similar intermolecular forces.

When solid solutes are dissolved in solvent, then following equilibrium exist.

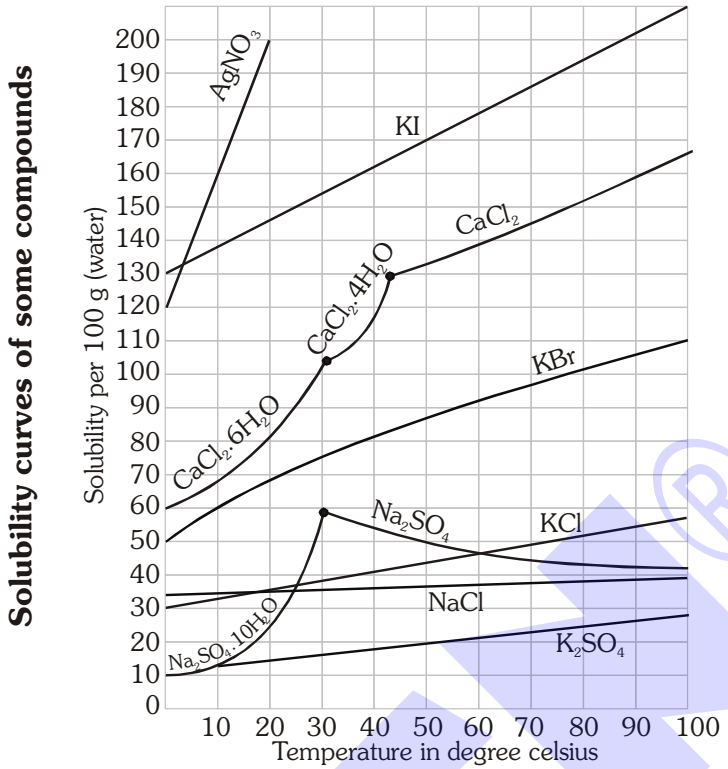
Solute + Solvent $\underset{\text{Crystallization}}{\overset{\text{Dissolution}}{\rightleftharpoons}}$ Solution. ; ΔH may be positive or negative

Such a solution in which no more solute can be dissolved at the same temperature and pressure is called a **saturated solution**. An unsaturated solution is one in which more solute can be dissolved at the same temperature. The solution which is in dynamic equilibrium with undissolved solute is the saturated solution and contains the maximum amount of solute dissolved in a given amount of solvent.

3.1 Effect of temperature :

The solubility of a solid in a liquid is significantly affected by temperature changes, obeying ***Le Chateliers Principle***. In general, if in a nearly saturated solution, the dissolution process is endothermic ($\Delta_{sol}H > 0$), the solubility should increase with rise in temperature and if it is exothermic ($\Delta_{sol}H < 0$) the solubility should decrease. These trends are also observed experimentally.

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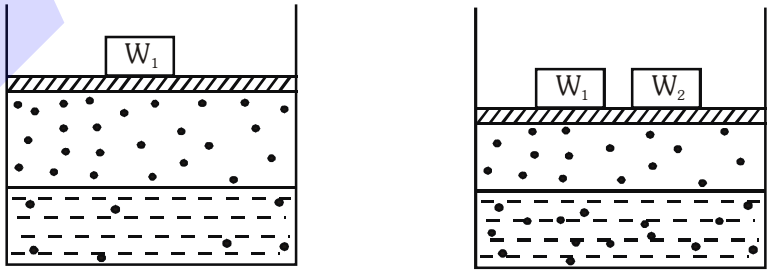
3.2 Effect of pressure :

Pressure does not have any significant effect on solubility of solids in liquids. It is so because solids and liquids are highly incompressible and practically remain unaffected by changes in pressure.

4. SOLUBILITY OF GASES IN LIQUID

Certain gases are highly soluble in water like NH_3, HCl, etc, and certain gases are less soluble in water like O_2, N_2, He, etc. Solubility of gases is greatly affected by pressure and temperature. Increasing pressure increases solubility and increase in temperature decreases solubility since dissolution of any gas in any liquid is exothermic in nature.

4.1 Henry' Law :



(a) Low pressure (b) High pressure

Effect of pressure on the solubility of a gas. The concentration of dissolved gas is proportional to the pressure on the gas above the solution.

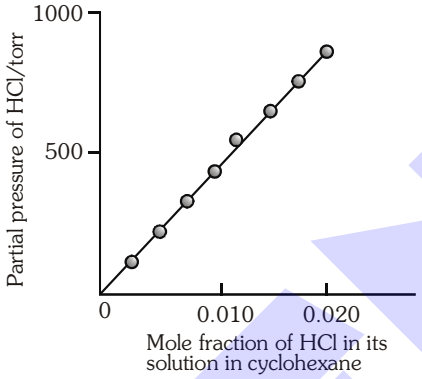
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The partial pressure of the gas in vapour phase (p) is proportional to the mole fraction of the gas (x) in the solution, at constant temperature

$$P = K_H X$$

K_H = Henry's Law Constant

* Henry's Law Constant depends on nature of gas and liquid as well as temperature.
* K_H increases with increases in temperature.
* Greater K_H means low solubility.



Experimental results for the solubility of HCl gas in cyclohexane at 293 K. The slope of the line is the Henry's Law constant K_H

Values of Henry's Law Constant for Some Selected Gases in Water

Gas	Temperature/K	K_H /kbar
He	293	144.97
H_2	293	69.16
N_2	293	76.48
N_2	303	88.84
O_2	293	34.86
O_2	303	46.82
Argon	298	40.3
CO_2	298	1.67
Formaldehyde	298	1.83×10^{-5}
Methane	298	0.413
Vinyl chloride	298	0.611

4.2 Limitations of Henry' Law :

(1) It is valid only for ideal behaviour of gas. As none of the gas is ideal, this law may be applied at low pressure and high temperature.

(2) It gives better result when the solubility of gas in the liquid is low.

(3) The gas should neither dissociate nor associate in the liquid.

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4.3 Henry' Law application :

(1) To increase the solubility of CO_2 in soft drinks and soda water, the bottle is sealed under high pressure.

(2) Scuba divers must cope with high concentrations of dissolved gases while breathing air at high pressure underwater. Increased pressure increases the solubility of atmospheric gases in blood. When the divers come towards surface, the pressure gradually decreases. This releases the dissolved gases and leads to the formation of bubbles of nitrogen in the blood. This blocks capillaries and creates a medical condition known as bends, which are painful and dangerous to life. To avoid bends, as well as, the toxic effects of high concentrations of nitrogen in the blood, the tanks used by scuba divers are filled with air diluted with helium (11.7% helium, 56.2% nitrogen and 32.1% oxygen).

(3) At high altitudes the partial pressure of oxygen is less than that at the ground level. This leads to low concentrations of oxygen in the blood and tissues of people living at high altitudes or climbers. Low blood oxygen causes climbers to become weak and unable to think clearly, symptoms of a condition known as anoxia.

4.4 Effect of Temperature :

Solubility of any gas in any liquid decreases with rise in temperature as dissolution is an exothermic process.

At constant pressure, $$\ell n\frac{C_2}{C_1}=\frac{\Delta H_{sol}}{R}\left(\frac{1}{T_1}-\frac{1}{T_2}\right)$$

where C = molar concentration of gas in solution

Ex.1. ***If N_2 gas is bubbled through water at 293 K, how many millimoles of N_2 gas would dissolve in 1 litre of water? Assume that N_2 exerts a partial pressure of 0.987 bar. Given that Henry's law constant for N_2 at 293 K is 76.48 kbar.***

Sol: $$x\ (Nitrogen)=\frac{p(nitrogen)}{K_H}=\frac{0.98\ bar}{76,480\ bar}=1.29\times10^{-5}$$

As 1 litre of water contains 55.5 mol of it, therefore if n represents number of moles of N_2 in solution,

$$x\ (Nitrogen)=\frac{n\,mol}{n\,mol+55.5\,mol}=\frac{n}{55.5}=1.29\times10^{-5}$$

(n in denominator is neglected as it is < < 55.5)

Thus n = 1.29 × 10^{-5} × 55.5 mol = 7.16 × 10^{-4} mol = 0.716 m mol

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Ex.2 ***The air is a mixture of a number of gases. The major components are oxygen and nitrogen with approximate proportion of 20% is to 79% by volume at 298 K. The water is in equilibrium with air at a pressure of 10 atm. At 298 K, if the Henry's law constants for oxygen and nitrogen are 3.30×10^7 mm and 6.51×10^7 mm respectively, calculate the composition of these gases in water.***

Sol: *Percentage of oxygen (O_2) in air = 20%*

Percentage of nitrogen (N_2) in air = 79%

Also, it is given that water is in equilibrium with air at a total pressure of 10 atm, that is, (10 × 760) mm Hg = 7600 mm Hg

Partial pressure of oxygen, $p_{O_2} = \frac{20}{100} \times 7600\,\text{mm Hg} = 1520$ *mm Hg*

Partial pressure of nitrogen, $p_{N_2} = \frac{79}{100} \times 7600\,\text{mm Hg} = 6004$ *mm of Hg*

Now, according to Henry's law :

$p = K_H x$

For oxygen :

$p_{O_2} = K_H . x_{O_2}$

$\Rightarrow x_{O_2} = \frac{p_{O_2}}{K_H} = \frac{1520\ \text{mm of Hg}}{3.30 \times 10^7\,\text{mm of Hg}}$ *(Given* $K_H = 3.30 \times 10^7$ *mm of Hg* $= 4.61 \times 10^{-5}$

For nitrogen,

$p_{N_2} = K_H . x_{N_2}$

$\Rightarrow x_{N_2} = \frac{P_{N_1}}{K_H} = \frac{6004\ \text{mm Hg}}{6.51 \times 10^7\ \text{mm Hg}} = 9.22 \times 10^{-5}$

Hence, the mole fractions of oxygen and nitrogen in water are 4.61 × 10^{-5} and 9.22 × 10^{-5} respectively.

5. Vapour pressure :

The Evaporation of a Liquid in a Closed Container

When a liquid is taken in a closed vessel at constant temperature, then there are two process which takes place.

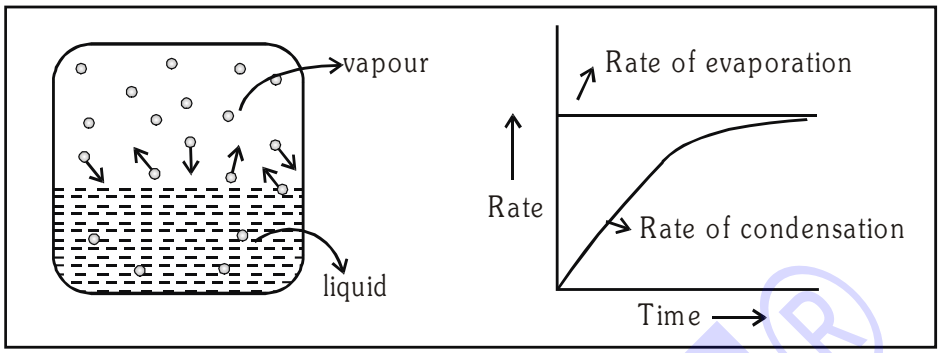
(i) Evaporation

(ii) Condensation

In the constant evaporation from the surface particles continue to break away from the surface of the liquid.

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As the gaseous particles bounce around, some of them will hit the surface of the liquid again, and will be trapped there. This is called condensation. The rate of condensation increases with time, but rate of evaporation remain constant. There will rapidaly be an equilibrium set up in which the number of particles leaving the surface is exactly balanced by the number rejoining it.



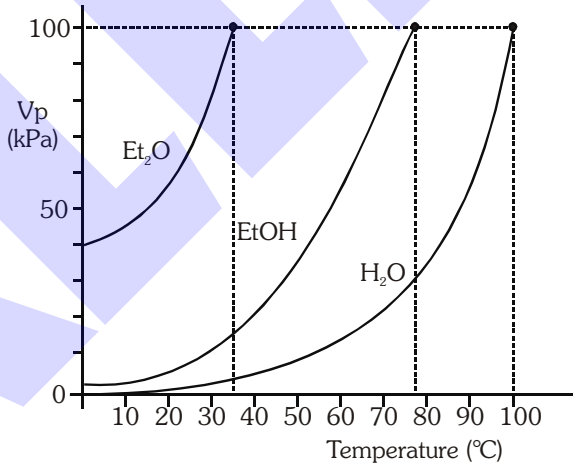
This pressure at equilibrium is called the **vapour pressure** (also known as **saturation vapour pressure**) **of the liquid at that temperature.**

5.1 Effect of Temperature on vapour pressure

When the space above the liquid is saturated with vapour particles, you have this equilibrium occuring on the surface of the liquid :

$$\text{liquid} \rightleftharpoons \text{vapour}, \quad \Delta H > 0$$

The forward change (liquid to vapour) is endothermic. It needs heat to convert the liquid into the vapour. According to Le Chatelier, increasing the temperature of a system in a dynamic equilibrium favours the endothermic change. That means increasing the temperature increase the amount of vapour present, and so increases the vapour pressure.



Effect of nature : $V.pr\ \alpha\ \frac{1}{\text{Inter molecular attraction force(I.M.A.F.)}}$

$n\text{–}C_6H_{14} > n\text{–}\ C_7H_{16} > C_8H_{18}$

The dependence of vapour pressure of a liquid on temperature is given by clausius-clapeyron equation : ln

$$\frac{P2}{P_1} = \frac{\Delta H_{vap.}}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

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5.2 Nature of liquid : Weaker the intermolecular attraction , higher will be the vapour pressure.

6. VAPOUR PRESSURE OF LIQUID SOLUTION

6.1 Raoult's law :

The partial pressure of any volatile constituents of a solution at a constant temperature is equal to the vapour pressure of pure constituents multiplied by the mole fraction of that constituent in the solution, at equilibrium.

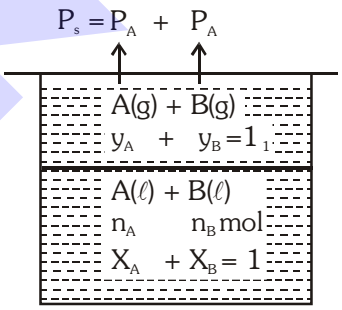
6.2 Vapour pressure of liquid – liquid solution :

Let P_A and P_B be the partial vapour pressures of two constituents A and B in solution and P_A^0 , P_B^0 are the vapour pressures in pure state respectively.

Thus, according to Raoult's law ,

$$P_A = \frac{n_A}{n_A + n_B} P_A^0 = X_A P_A^0 \quad ....(1)$$

and $$P_B = \frac{n_B}{n_A + n_B} P_B^0 = X_B P_B^0 \quad ....(2)$$



If total pressure be P_S, then

$$P_S = P_A + P_B = X_A P_A^0 + X_B P_B^0$$

$$P_S = X_A P_A^0 + (1 - X_A) P_B^0 \qquad [\because X_A + X_B = 1]$$

$$P_S = X_A P_A^0 - X_A P_B^0 + P_B^0$$

$$P_S = X_A [P_A^0 - P_B^0] + P_B^0$$

6.2.1 Relation between Dalton's Law and Raoult's Law :

The composition of the vapour in equilibrium with the solution can be calculated applying Dalton's law of partial pressures. Let the mole fraction of A and B in vapours be Y_A and Y_B respectively.

$$p_A = Y_A P_S = X_A P_A^0 \quad .....(1)$$

$$p_B = Y_B P_S = X_B P_B^0 \quad .....(2)$$

Now, $$X_A = \frac{Y_A \cdot P_S}{P_A^0} \quad \text{and} \quad X_B = \frac{Y_B \cdot P_S}{P_B^0}$$

As, $$X_A + X_B = 1, \ \frac{Y_A \cdot P_S}{P_A^0} + \frac{Y_B \cdot P_S}{P_B^0} = 1$$

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$$\therefore \quad \frac{1}{P_S} = \frac{Y_A}{P_A^0} + \frac{Y_B}{P_B^0}$$

Hence, the total vapour pressure of solution may be calculated from liquid composition at equilibrium as

$$P_S = X_A \cdot P_A^0 + X_B \cdot P_B^0$$

and from vapour composition at equilibrium as

$$\frac{1}{P_S} = \frac{Y_A}{P_A^0} + \frac{Y_B}{P_B^0}$$

6.2.2 Comparision between liquid and vapour composition :

$$\frac{Y_A}{Y_B} = \frac{P_A / P_S}{P_B / P_S} = \frac{P_A}{P_B} = \frac{X_A \cdot P_A^0}{X_B \cdot P_B^0}$$

If A is more volatile than B $\left(P_A^0 > P_B^0\right)$, then $\frac{Y_A}{Y_B} > \frac{X_A}{X_B}$

It means that the mole-fraction of A in vapour form is relatively greater than that in liquid form, relative to B. Hence, in any ideal solution, vapour is always more richer in the more volatile component, relative to liquid.

6.2.3 Raoult's law as a special case of Henry's law :

From Raoult's law, the vapour pressure of volatile component in the solution is $P = P^0.X$.

In the solution of gas in liquid, one component is so volatile that it exist as gas and its pressure is given by Henry's law as $P = K_H.X$

In both laws, the partial pressure of volatile component is directly proportional to its mole-fraction in solution. Only the proportionality constant K_H differs from P^0. Hence, Raoult's law becomes a special case of Henry's law in which K_H becomes P^0.

6.3 Vapour pressure of solution of solids in liquid :

Let us assume A = non volatile solid & B = volatile liquid

According to Raoult's law –

$\because \quad P_s = X_A P_A^{\ 0} + X_B P_B^{\ 0}$

for A, $P_A^{\ 0} = 0$

$\therefore \quad P_s = X_B P_B^{\ 0}$ ....(5)

Let $P_B^{\ 0} = P^0$ = Vapour pressure of pure state of solvent.

here X_B is mole fraction of solvent

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$$P_S = \frac{n_B}{n_A + n_B} P^0 \qquad ....(6)$$

$$P_S \propto \frac{n_B}{n_A + n_B}$$ i.e. vapour pressure of solution $\propto$ mole fraction of solvent

$\Rightarrow \quad P_S = X_B P_B^0 \Rightarrow P_S = (1 - X_A) P_B^0$

$\Rightarrow \quad P_S = P_B^0 - X_A P_B^0$

$\Rightarrow \quad \frac{P_B^0 - P_S}{P_B^0} = X_A$

or $\quad \frac{P^0 - P_S}{P^0} = X_A \qquad ....(7)$;

$$\boxed{\frac{\mathbf{P^0 - P_S}}{\mathbf{P^0}} = \frac{\mathbf{n_A}}{\mathbf{n_A + n_B}}} \qquad ....(8)$$

or $\quad \frac{P^0}{P^0 - P_S} = \frac{n_A + n_B}{n_A}$

or $\frac{P^0}{P^0 - P_S} = 1 + \frac{n_B}{n_A}$ or $\frac{P^0}{P^0 - P_S} - 1 = \frac{n_B}{n_A}$

$$\frac{P_S}{P^0 - P_S} = \frac{n_B}{n_A}$$

$$\boxed{\frac{\mathbf{P^0 - P_S}}{\mathbf{P_S}} = \frac{\mathbf{n_A}}{\mathbf{n_B}} = \frac{\mathbf{w_A . m_B}}{\mathbf{m_A . w_B}}} \qquad ....(9)$$

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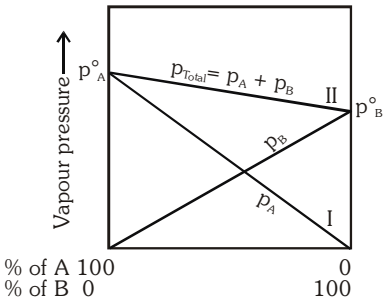
6.4 Ideal and Non ideal solutions :

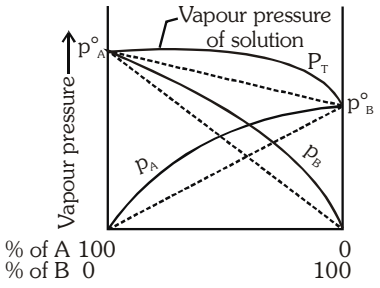
The solutions which obey Raoult's law over the entire range of concentration are known as ideal solutions.

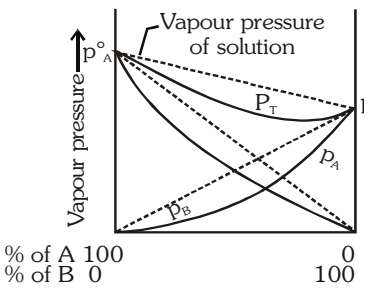
Table : Comparison between Ideal and Non-ideal solutions

	Ideal solutions	Non-ideal solutions	
		+ve deviation from Raoult's law	– deviation from Raoult's law
1.	Obeys Raoult's law at every concentration.	Do not obey Raoult's law.	Do not obey Raoult's law.
2.	$\Delta H_{mix} = 0$; Neither heat is evolved nor absorbed during dissolution.	$\Delta H_{mix} > 0$. Endothermic dissolution; heat is absorbed.	$\Delta H_{mix} < 0$; exothermic dissolution heat is evolved.
3.	$\Delta V_{mix} = 0$; total volume of solution is equal to sum of volumes of the components.	$\Delta V_{mix} > 0$. Volume is increased after dissolution.	$\Delta V_{mix} < 0$; volume is decreased during dissolution.
4.	$P = p_A + p_B = p_A^0 X_A + p_B^0 X_B$ i.e., $P_A = p_A^0 X_A ; p_B = p_B^0 X_B$	$P_A > p_A^0 X_A ;\ p_B > p_B^0 X_B$ $\therefore (P_A + p_B) > (p_A^0 X_A + p_B^0 X_B)$	$P_A < p_A^0 X_A ;\ p_B < p_B^0 X_B$ $\therefore (P_A + p_B) < (p_A^0 X_A + p_B^0 X_B)$
5.	A — A, A — B, B — B interactions should be same, i.e., 'A' and 'B' are identical in shape, size and character.	A — B, attraction force should be weaker than A — A and B — B attractive forces. 'A' and 'B' have different shape, size and character.	A — B, attraction force should be greater than A — A and B — B attractive forces. 'A' and 'B' have different shape, size and charater.
6.	Escaping tendency of 'A' and 'B' should be same in pure liquids and in the solution.	'A' and 'B' escape easily showing high vapour pressure than the expected value.	Escaping tendency of both components A and B is lowered showing lower vapour pressure than expected ideally.
	Example : dilute solutions ; benzene + toluene ; n-hexane + n-heptane ; chlorobenzene + bromobenzene ; ethyl bromide + ethyl iodide ; n-butyl chloride + n-butyl bromide $CCl_4 + SiCl_4$; $C_2H_4Br_2 + C_2H_4Cl_2$ $C_2H_5Br + C_2H_5Cl$	**Example :** acetone + ethanol ; acetone + CS_2 ; water + methanol ; water + ethanol ; $CCl_4 + CHCl_3$; CCl_4 + toluene ; acetone + benzene $CCl_4 + CH_3OH$; cyclohexane + ethanol	**Example :** acetone + aniline ; acetone + chloroform ; $CH_3OH + CH_3COOH$; $H_2O + HNO_3$; water + HCl ; acetic acid + pyridine ; $HNO_3 + CHCl_3$

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Ex.3. ***1 mole heptane (V.P. = 92 mm of Hg) is mixed with 4 mole Octane (V.P. = 31 mm of Hg), form an ideal solution. Find out the vapour pressure of solution.***

Sol. *Total mole = 1 + 4 = 5*

Mole fraction of heptane $= X_A = 1/5$

Mole fraction of octane $= X_B = 4/5$

$$P_S = X_A P_A^{\,0} + X_B P_B^{\,0} = \frac{1}{5}\times 92 + \frac{4}{5}\times 31 = 43.2 \text{ mm of Hg.}$$

Ex.4. ***At 88°C, benzene has a vapour pressure of 900 torr and toluene has a vapour pressure of 360 torr. What is the mole fraction of benzene in the mixture with toluene that will be boil at 88°C at 1 atm pressure, benzene – toluene form an ideal solution.***

Sol. $P_S = 760$ *torr, because solution boils at 88°C*

$\therefore\ 760 = 900\,x + 360\,(1 - x)$

$x = 0.74$ *where 'x' is mole fraction* C_6H_6.

Ex.5. ***The vapour pressure of benzene at 90°C is 1020 torr. A solution of 5 g of a solute in 58.5 g benzene has vapour pressure 990 torr. What is the molecular weight of solute ?***

Sol. $$\frac{P^0 - P_S}{P_S} = \frac{w \times M}{m \times W} \Rightarrow \frac{1020 - 990}{990} = \frac{5 \times 78}{m \times 58.5} \Rightarrow m = 220$$

Ex.6. ***Two liquids A and B form an ideal solution. At 300 K, the vapour pressure of a solution containing 1 mole of A and 3 mole of B is 550 mm of Hg. At the same temperature, if one more mole of B is added to this solution, the vapour pressure of the solution increases by 10 mm of Hg. Determine the vapour pressure of A and B in their pure states.***

Sol. *Let the vapour pressure of pure A be* $= p_A^0$ *; and the vapour pressure of pure B be* $= p_B^0$.

Total vapour pressure of solution (1 mole A + 3 mole B)

$= X_A.p_A^0 + X_B.p_B^0$ *[*X_A *is mole fraction of A and* X_B *is mole fraction of B]*

$$550 = \frac{1}{4}p_A^0 + \frac{3}{4}p_B^0 \quad \text{or} \quad 2200 = p_A^0 + 3p_B^0 \qquad .....(i)$$

Total vapour pressure of solution (1 mole A + 4 mole B) $= \frac{1}{5}p_A^0 + \frac{4}{5}p_B^0$

$$560 = \frac{1}{5}p_A^0 + \frac{4}{5}p_B^0$$

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$2800 = p_A^0 + p_B^0$ .....(ii)

Solving eqs. (i) and (ii)

p_B^0 = *600 mm of Hg = vapour pressure of pure B*

p_A^0 = *400 mm of Hg = vapour pressure of pure A*

Ex.7. ***Liquids 'A' and 'B' form an ideal solution. Calculate the vapour pressure of solution having 40 mole-percent of A in the vapour at equilibrium.*** $\left(P_A^0 = 80\text{cm Hg},\ P_B^0 = 30\text{cm Hg}\right)$

Sol. $$\frac{1}{P_{total}} = \frac{Y_A}{P_A^0} + \frac{Y_B}{P_B^0} = \frac{0.4}{80} + \frac{0.6}{30} = \frac{1}{40}$$

P_{total} = *40 cm Hg*

Ex.8. ***Liquids 'A' and 'B' form an ideal solution. Calculate the molar-fraction of 'A' in vapour form above the liquid solution containing 25 mole-percent of 'A' at equilibrium*** $\left(P_A^0 = 0.2\text{atm},\ P_B^0 = 0.5\text{atm}\right)$

Sol. $$Y_A = \frac{P_A}{P_{total}} = \frac{X_A \cdot P_A^0}{X_A \cdot P_A^0 + X_B \cdot P_B^0} = \frac{0.25 \times 0.2}{0.25 \times 0.2 + 0.75 \times 0.5} = \frac{2}{17}$$

Ex.9. ***Liquids 'A' and 'B' form an ideal solution. At 80°C,*** $P_A^0 = 0.4\text{bar}$ ***and*** $P_B^0 = 0.8\text{bar}$***. All the vapour above the liquid solution containing equal moles of both the liquids at equilibrium is collected in another empty vessel and condensed. Now, the condesate is heated to 80°C and all the vapours above the liquid solution at equilibrium is again collected in another empty vessel and condensed. What is the mole-fraction of 'B' is new condensate ?***

Sol. *For the first condensate,*

$$\frac{n_B'}{n_A'} = \frac{X_B'}{X_A'} = \frac{Y_B}{Y_A} = \frac{X_B}{X_A} \cdot \frac{P_A^0}{P_B^0} = \frac{n_B}{n_A} \times \frac{P_B^0}{P_A^0}$$

For second condensate,

$$\frac{n_B''}{n_A''} = \frac{X_B''}{X_A''} = \frac{Y_B'}{Y_A'} = \frac{X_B' \cdot P_B^0}{X_A' \cdot P_A^0} = \frac{n_B}{n_A} \times \left(\frac{P_B^0}{P_A^0}\right)^2 = \frac{x}{x} \times \left(\frac{0.8}{0.4}\right)^2 = \frac{4}{1}$$

$\therefore$ *Mole fraction of* $B = \frac{4}{5} = 0.8$

Note : For multi-step condensation at constant temperature,

$$\boxed{\frac{n_B^f}{n_A^f} = \frac{n_B^i}{n_B^i} \cdot \left(\frac{P_B^0}{P_A^0}\right)^n}$$ *n : number of steps*

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Ex.10. ***Liquid A & B form an ideal solution. In a cylinder piston arrangement, 2 moles of vapours of liquid A & 3 moles of vapours of liquid B are taken at 0.3 atm.*** $P_A^0 = 0.4\,atm$, $P_B^0 = 0.6\,atm$.

(i) Predict whether vapours will condense or not ?

(ii) If the vapours are compressed slowly & isothermly, at what pressure Ist drop of liquid will form

(iii) If the initial volume of vapours was 10L, at what volume Ist drop of liquid will form ?

(iv) What is the composition of Ist drop of liquid formed ?

(v) If the vapours are further compressed slowly & isothermally, at what P almost complete condensation will occur ?

(vi) What is the composition of last traces of vapours remained ?

(vii) What is the composition of system at 0.58 atm ?

(viii) What is the composition of system at 0.51 atm ? Also calculate moles of A & B in liquid & vapour form.

(ix) At what P, half of the total amount of vapours will condense ?

Sol. *(i)* $\frac{1}{P_T} = \frac{Y_A}{P_A^0} + \frac{Y_B}{P_B^0} = \frac{2/5}{0.4} + \frac{3/5}{0.6} \Rightarrow P_{total} = 0.5\ atm > 0.3\ atm$

$\therefore$ *100% gas, no condensation*

(ii) 0.5 atm.

(iii) $P_iV_i = P_fV_f$

$V_f = \frac{0.3 \times 10}{0.5} = 6\,L$

(iv) $P_{total} = X_A \cdot P_A^0 + X_B \cdot P_B^0$

or $0.5 = X_A \times 0.4 + (1 - X_A) \times 0.6 \Rightarrow X_A = 0.5$

(v) $P_T = 0.4 \times 0.4 + 0.6 \times 0.6 = 0.52\ atm$

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Note : *For a pure liquid, there is a fixed P(V.P.) below & above which, the system will be 100% gas & 100% liquid, respectively but for a solution, a pressure range exist in which both physical states will be present.*

(vi) $Y_A = \frac{X_A \cdot P_A^0}{P_{total}} = \frac{0.4 \times 0.4}{0.52} = \frac{4}{13}$

(vii) *100% liquid, A = 2mole and B = 3 mole*

(viii) $P_{total} = X_A P_A^0 + X_B P_B^0$

$0.51 = X_A \times (0.4) + (1 - X_A) \times 0.6$ & $Y_A = \frac{X_A P_A^0}{P_{total}} = \frac{0.45 \times 0.4}{0.51} = 0.35$

$X_A = 0.45$

A = 2mol
B = 3 mol

Liquid	Vapour
A = x mol	A = (2 – x)
B = y mol	B = (3 – y)

$X_A = \frac{9}{20} = \frac{x}{x+y}$ *and* $Y_A = \frac{18}{51} = \frac{2-x}{(2-x)+(3-y)}$

$\therefore$ $x = 1.09$; $y = 1.33$

(ix)

A = 2mole
B = 3 mole

Liquid = 2.5 mole	Vapour = 2.5 mole
A = z mole	A = (2 – z) mole
B = (2.5 – z) mole	B = (0.5 – z) mole

$P_T = \frac{z}{2.5} \times 0.4 + \frac{2.5 - z}{2.5} \times 0.6$

$\frac{2-z}{2.5} = \frac{z \times 0.4}{2.5 \times P_T}$

On solving, z = 1.12, P_T = 0.5104 atm

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6.5 Graphs for ideal Binary solution of liquid A & liquid B :

(Assume $P_A^0 < P_B^0$)

I. Vapour pressure V/s liquid composition

$$P_A = X_A P_A^0 = (1 - X_B) P_A^0$$

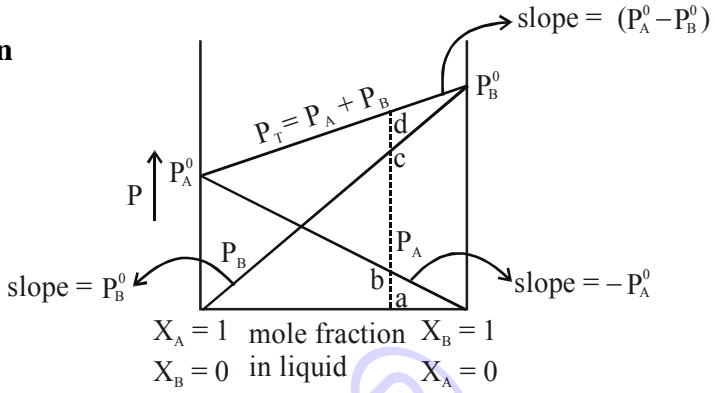
$$P_A = P_A^0 - X_B P_A^0$$

$$P_B = X_B P_B^0$$

$$P_T = X_A P_A^0 + X_B P_B^0 = (1 - X_B) P_A^0 + X_B P_B^0$$

$$\therefore \quad P_T = P_A^0 + X_B \left(P_B^0 - P_A^0 \right)$$

As $P_T = P_A + P_B$, $\quad ad = ab + ac$



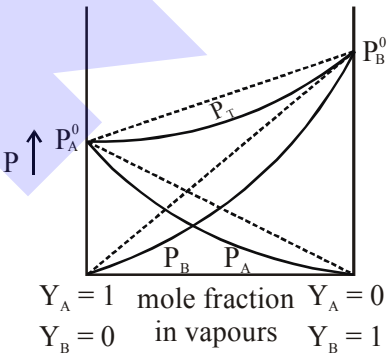
II. Vapour pressure V/s vapour composition

$$\frac{1}{P_T} = \frac{Y_A}{P_A^0} + \frac{Y_B}{P_B^0}$$

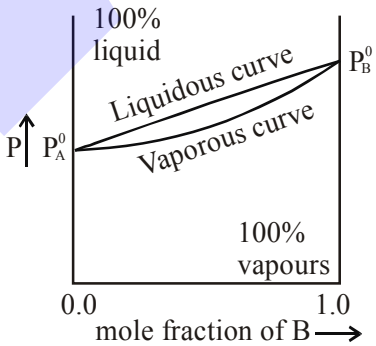
$$\frac{1}{P_T} = \frac{1}{P_A^0} + Y_B \left(\frac{1}{P_B^0} - \frac{1}{P_A^0} \right)$$

$$\frac{1}{y} = c + mx$$

So curve will be rectangular hyperbola.



III. Vapour pressure V/s composition



(i) The V.P. of ideal solution always lie in between the V.P. of pure components.

(ii) Below vaporous curve, the system will be 100% vapour & above liquidous curve, 100% liquid. Both the physical states exist only in between the curves.

(iii) At any composition, the physical state of system may be changed by changing the pressure.

(iv) At any pressure in between P_A^0 and P_B^0, the physical state of system may be changed by changing the composition.

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6.6 Boiling point :

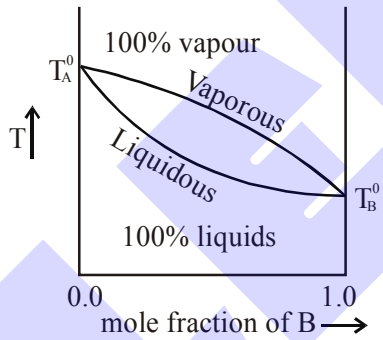
Boiling point of a liquid is the temperature at which the vapour pressure of the liquid become equal to the atmospheric pressure.

Ex.11 ***Liquid 'A' and 'B' form an ideal solution. At 27°C, the vapour pressure of pure liquids 'A' and 'B' are 0.6 atm and 1.2 atm, respectively. What the composition of liquid solution boiling at 27°C?***

Sol. $P_T = X_A \cdot P_A^0 + X_B \cdot P_B^0$

$$or\ 1 = X_A \times 0.6 + (1 - X_A) \times 1.2 \quad \Rightarrow \quad X_A = \frac{1}{3}$$

6.7 Boiling point curves for ideal binary solution :



(i) The boiling point of ideal solution always lie in between the boiling points of pure liquids.

(ii) Below liquid curve, the system in 100% liquid and above vapour curve, the system is 100% vapour. Both the physical states exists only in between the curves.

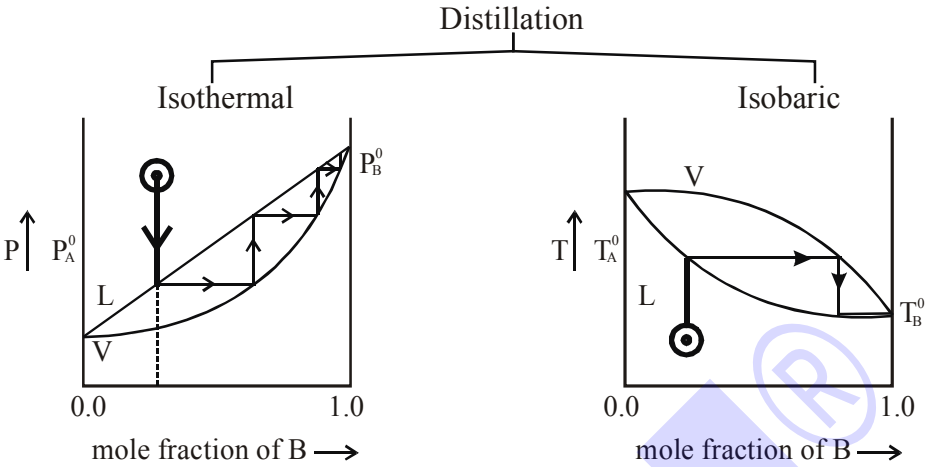
(iii) At any composition, the physical state of system may be changed by changing the temperature.

(iv) At any temperature in between T_A^0 and T_B^0, the physical state of system may be changed by changing the composition.

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6.8 Distillation

It is the method of seperation of liquids by converting them into vapours (boiling).

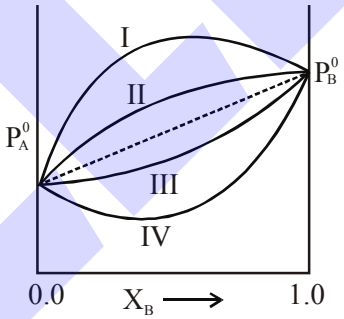


(i) The separation of liquid by distillation occurs because at any T or P, the composition of distillate or condensate is different than the composition of original liquid.

(ii) With the elimination of vapour above the liquid, the boiling point of residual liquid increases.

(iii) The boiling point of distillate is less than that of original liquid.

6.9 Graphs for Non-ideal solutions and azeotropic mixture



(I) Large positive deviation : $(V.P.)_{solution} > P_B^0$ at some composition

(II) Small positive deviation : $P_A^0 < (V.P.)_{solution} < P_B^0$

(III) Small negative deviation : $P_A^0 < (V.P.)_{solution} < P_B^0$

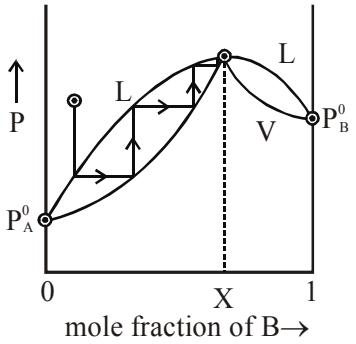
(IV) Large negative deviation : $(V.P.)_{solution} < P_A^0$ at some composition

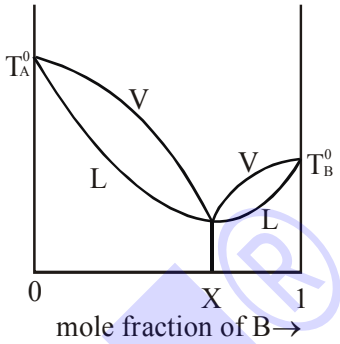
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6.9.1 Konowaloff's rule :

In ideal or non-idal solution the vapour is always more rich in the component, addition of which in liquid, increase the vapour pressure of solution.

6.9.2 Large (+)ve deviation (Minimum boiling azeotrope)





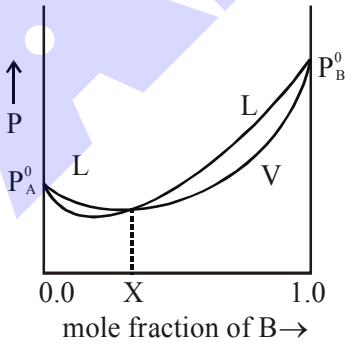
The solution of large positive deviation can not be seperated by distillation because at composition 'x', the liquid & vapour composition becomes identical.

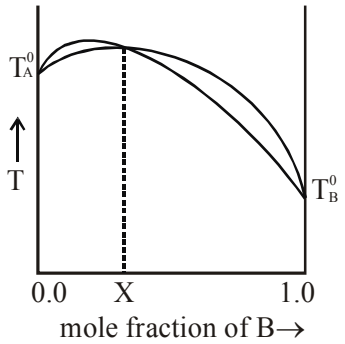
Such solution which can not be saperated by distillation are called **azeotropic mixture or constant boiling solution.**

Minimum boiling point Azeotropic

Mixture	% Composition of azeotrope	Boiling point (pressure = 1atm)
1. Water-Ethanol	96 Ethanol	78.15°C
2. Pyridine-Water	57.00 Pyridine	92.60°C
3. Ethanol-Benzene	32.4 Ethanol	67.80°C
4. Acetic acid-Toluene	28.0 Acetic acid	105.40°C

5.9.3 Large (–)ve deviation (Maximum boiling azeotrope)





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Maximum boiling point Azeotropic

Mixture	% Composition of azeotrope	Boiling point (pressure = 1atm)
1. Nitric acid-Water	68% Nitric acid	125.5℃
2. Acetic acid-Pyridine	65% Pyridine	139.0℃
3. Chloroform-Acetone	80% Chloroform	65.0℃
4. Hydrogen chloride-Water	79.8 Water	108.6℃

Note : Azeotrope is not formed in ideal solution or solution of small deviations.

7. COLLIGATIVE PROPERTIES

Properties of a solution which depends on the number of solute particles irrespective of their nature, relative to the total number of particles present in solution are called colligative properties.

The following properties are colligative properties of solution :

(i) Relative lowering of vapour pressure.

(ii) Elevation in boiling point.

(iii) Depression in freezing point.

(iv) Osmotic pressure.

7.1 Lowering of vapour pressure :

When a non-volatile solute 'A' is dissolved in a pure solvent 'B', the vapour pressure of the solvent is lowered i.e. the vapour pressure of a solution is always lower than that of pure solvent, because the escaping tendency of solvent molecules decreases.

If at a certain temperature P° is the vapour pressure of pure solvent, and P_s is the vapour pressure of solution then

Lowering of vapour pressure = $P^\circ - P_s$

Relative lowering of vapour pressure = $\frac{P^0 - P_s}{P^0}$

from equation (8)

$$\frac{\mathbf{P^0 - P_S}}{\mathbf{P^0}} = \frac{\Delta \mathbf{P}}{\mathbf{P^0}} = \frac{\mathbf{n_A}}{\mathbf{n_A + n_B}} = \mathbf{X_A}$$

For a very dilute solution $n_A << n_B$

so $$\frac{\mathbf{P^0 - P_S}}{\mathbf{P^0}} = \frac{\mathbf{n_A}}{\mathbf{n_B}} = \frac{\mathbf{w_A}}{\mathbf{m_A}} \times \frac{\mathbf{m_B}}{\mathbf{w_B}}$$

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Ex.12. Calculate the vapour pressure lowering caused by addition of 50 g of sucrose (molecular mass = 342) to 500 g of water if the vapour pressure of pure water at 25°C is 23.8 mm Hg.

Sol. According to Raoult's law,

$$\frac{p_0 - p_s}{p_0} = \frac{n}{n+N} \quad \text{or} \quad \Delta p = \frac{n}{n+N}.p_0$$

Given : $n = \frac{50}{342} = 0.146$; $N = \frac{500}{18} = 27.78$ and $p_0 = 23.8$ mmHg

Substituting the values in the above equation,

$$\Delta p = \frac{0.146}{0.146 + 27.78} \times 23.8 = 0.124 \text{ mm Hg}$$

7.1.1 Measurement of Lowering in Vapour Pressure by Dynamic Method (Ostwald and Walker Method)

The apparatus used is shown in Fig. It consists of two sets of bulbs. The first set of three bulbs is filled with solution to half of their capacity and second set of another three bulbs is filled with the pure solvent.

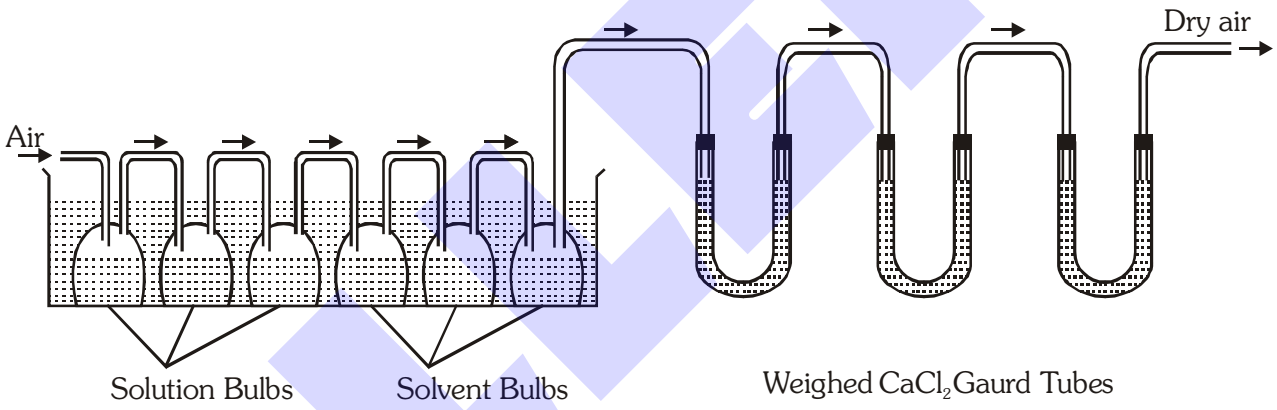


Fig. Ostwald and Walker method

Each set is separately weighed accurately. Both sets are connected to each other and then with the accurately weighed set of guard tubes filled with anhydrous calcium chloride or some other dehydrating agents like P_2O_5, conc. H_2SO_4 etc. The bulbs of solution and pure solvent are kept in a thermostat maintained at a constant temperature.

A current of pure dry air is bubbled through the series of bulbs as shown in fig. The air gets saturated with the vapour in each set of bulbs. The air takes up an amount of vapours proportional to the vapour pressure of the solution first and then it takes up more amount of vapours from the solvent which is proportional to the difference in the vapour pressure of the solvent and the vapour pressure of solution, i.e., $p_0 - p_s$.

The two sets of bulbs are weighed again. The guard tubes are also weighed.

Loss in mass in the solution bulbs $\propto p_s$

Loss in mass in the solvent bulbs $\propto (p_0 - p_s)$

Total loss in both sets of bulbs $\propto [p_s + (p_0 - p_s)] \propto p_0$

Total loss in mass of both sets of bulbs is equal to gain in mass of guard tubes.

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Thus, $\frac{p_0 - p_s}{p_0} = \frac{\text{Loss in mass in solvent bulbs}}{\text{Total loss in mass in both sets of bulbs}} = \frac{\text{Loss in mass in solvent bulbs}}{\text{Gain in mass of guard tubes}}$

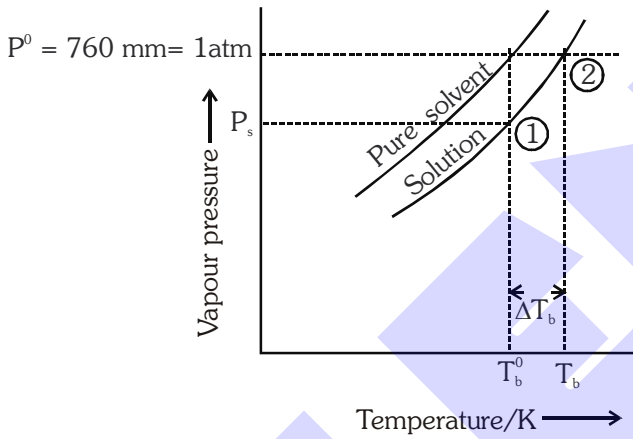
Further, we know from Raoult's law,

$$\frac{p_0 - p_s}{p_0} = \frac{w_A / m_A}{w_A / m_A + w_B / m_B}$$

$$\therefore \quad \frac{\text{Loss in mass of solvent bulbs}}{\text{Gain in mass of gaurd tubes}} = \frac{w_A / m_A}{w_A / m_A + w_B / m_B}$$

6.2 Elevation in boiling point (Ebullioscopy) :

$\text{solvent}(\ell) \rightleftharpoons \text{solvent}(g)$



The vapour pressure curve for solution lies below the curve for pure solvent .

ΔT_b denotes the elevation of boiling point of a solution.

When a non-volatile solute A is dissolved in a pure solvent B, its vapour pressure decreases and hence the boiling point increases. The difference ΔT_b of boiling points of the solution and pure solvent is called elevation in boiling point.

If T_b^0 is the boiling point of pure solvent and T_b is the boiling point of the solution then, $T_b > T_b^0$

and the elevation in boiling point $\Delta T_b = T_b - T_b^0$

Experiments have shown that for dilute solutions, the elevation of boiling point (ΔT_b) is directly proportional to the molal concentration of the solute in a solution. Thus

$$\Delta T_b \propto m$$

or $$\Delta T_b = K_b.m$$

where K_b = boiling point elevation constant or molal elevation constant or ebulioscopic constant.

Molal elevation constant is characteristic of a particular solvent and can be calculated from the thermodynamical relationship.

$$\mathbf{K_b = \frac{RT_b^2.M}{1000\ \Delta H_{vap}}}$$

Where, R is molar gas constant = 2 cal/mol–K , M = molar mass of solvent

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T_b is the boiling point of the pure solvent (in K)

and ΔH_{vap} is the latent heat of vaporisation of pure solvent

For water $K_b = \frac{2 \times (373)^2}{1000 \times 540} = 0.515$ K-kg/mol

The molal elevation constant for some common solvents are given in the following table

Solvent	B.P. (°C)	Molal elevation constant
Water	100.0	0.52
Acetone	56.0	1.70
Chloroform	61.2	3.63
Carbon tetrachloride	76.8	5.03
Benzene	80.0	2.53
Ethyl alcohol	78.4	1.20

Ex.13. ***0.15 g of a substance dissolved in 15 g of solvent boiled at a temperature higher by 0.216°C than that of the pure solvent. What is the molecular weight of the substance. [K_b for solvent = 2.16 K-kg/mol]***

Sol. ***Given : $K_b = 2.16°C$, $w = 0.15$ g, $\Delta T_b = 0.216°C$, $W = 15$ g***

ΔT_b = molality × K_b

$$\Delta T_b = \frac{w}{m \times W} \times 1000 \times K_b$$

$$0.216 = \frac{0.15}{m \times 15} \times 1000 \times 2.16$$

$$m = \frac{0.15 \times 1000 \times 2.16}{0.216 \times 15} = 100$$

Ex.14 ***The rise in boiling point of a solution containing 1.8 g glucose in 100 g of a solvent is 0.1°C. The molal elevation constant of the liquid is :–***

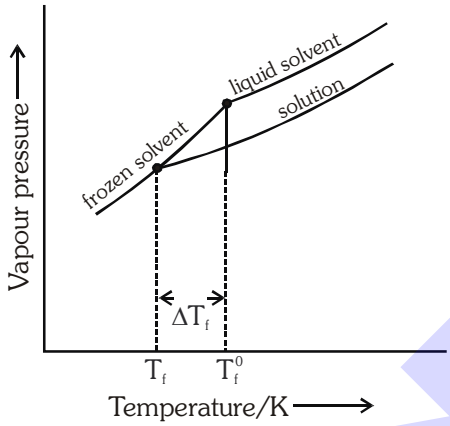
Sol. *$\Delta T_b = 0.1°C$, $m = 180$, $W = 100$, $w = 1.8$*

$$K_b = \frac{\Delta T_b \times m \times W}{1000 \times w} = \frac{180 \times 0.1 \times 100}{1000 \times 1.8} = 1.0$$

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7.3 Depression in freezing point (Cryoscopy) :

The freezing point of a liquid is that temperature at which the liquid and its solid state exist in equilibrium with each other. It may also be defined as the temperature at which the liquid and solid states of a substance have the same vapour pressure.



When a non-volatile non-electrolyte is dissolved in a pure solvent, the vapour pressure of the solvent is lowered and it become equal to that of solid solvent at lower temperature.

If T_f^0 is the freezing point of pure solvent and (T_f) is the freezing point of its solution then,

$$T_f < T_f^0$$

The difference in the freezing point of pure solvent and solution is the depression of freezing point (ΔT_f) Thus,

$$T_f^0 - T_f = \Delta T_f$$

Similar to elevation in boiling point, depression in freezing point for dilute solution is directly proportional to its molality,

$$\Delta T_f \propto m$$

or $$\Delta T_f = K_f . m$$

where K_f is called freezing point depression constant or molal depression constant or cryoscopic constant.

K_f is characteristic of a particular solvent and can be calculated from the thermodynamical relationship

$$K_f = \frac{R \cdot T_f^{0^2} \cdot M}{1000 \cdot \Delta H_{fus}}$$

where, T_f^0 is the freezing point of pure solvent in (Kelvin) and ΔH_{fus} is the latent heat of fusion pure solvent.

For water,

$$K_f = \frac{0.002 \times (273)^2}{80} = 1.86 \text{ K-kg/mol}$$

The molal depression constant for some common solvents are given in the following table

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Solvent	F.P.(°C)	Molal depression solvents
Water	0.0	1.86
Ethyl alcohol	– 114.6	1.99
Chloroform	– 63.5	4.79
Carbon tetrachloride	– 22.8	31.8
Benzene	5.5	5.12
Camphor	179.0	39.70

Ex.15. ***If freezing point of a solution prepared from 1.25 g of a non electrolyte and 20 g of water is 271.9 K, the molar mas of the solute will be – (K_f of water = 1.86 K-kg/mol)***

Solution : *Given* $T_f = 271.9\ K$

$w = 1.25\ g \qquad W = 20\ g \quad K_f = 1.86$

$\Delta T_f = T_f^0 - T_f = 273 - 271.9 = 1.1\ K$

$\Delta T_f = molality \times K_f$

$$\Rightarrow \quad \Delta T_f = \frac{w}{m \times W} \times 1000 \times K_f$$

$$or \quad m = \frac{w \times 1000 \times K_f}{\Delta T_f \times W} = \frac{1.25 \times 1000 \times 1.86}{1.1 \times 20} = 105.68\ mol/kg$$

Ex.16. ***Molal depression constant for water is 1.86. What is freezing point of a 0.05 molal solution of a non electrolyte in water ?***
(K_f of water = 1.86 K-kg/mol)

Solution : $\Delta T_f = molality \times K_f$

$= 0.05 \times 1.86 = 0.093°C$

$T_f = T_f^0 - 0.093 = 0 - 0.093$

$T_f = -0.093°C$

Ex.17. ***2m aqueous urea solution is cooled to –7.44°C. Calculate the mass percent of water present in solution, which will separate as ice, (K_f of water = 1.86 K-kg/mol)***

Sol. *Let the initial mass of water in the solution = 1kg*

$\therefore$ *Moles of solute = 2*

$$Now,\ \Delta T_f = K_f m = K_f . \frac{n_{solute}}{Kg_{solvent}} \quad or\ 7.44 = 1.86 \times \frac{2}{Kg_{solvent}}$$

$\therefore$ $Kg_{solvent}$ *left in the solution = 0.5*

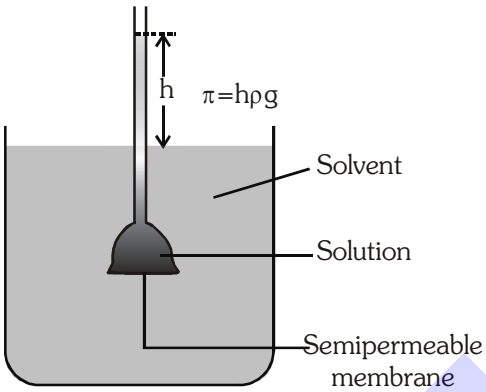
$$\therefore \quad Mass\ percent\ of\ water\ separated\ as\ ice = \frac{0.5}{1} \times 100 = 50\%$$

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7.4 Osmosis and osmotic pressure :

7.4.1 Osmosis :

Osmosis is defined as the spontaneous flow of solvent molecules through semipermeable membrane from a pure solvent to solution.



Level of solution rises in the funnel due to osmosis of solvent

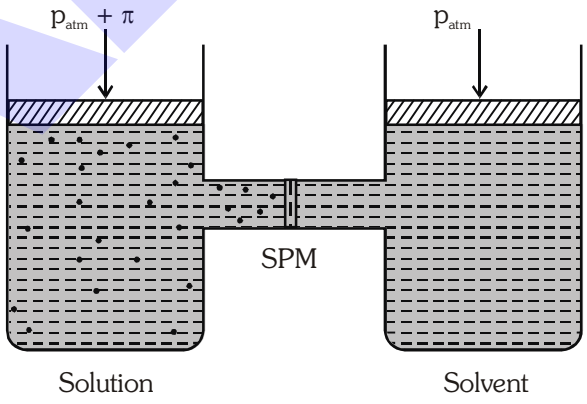
7.4.2 Semi-permeable membrane (SPM) :

There are the membranes (substances) which allow selective movement of particles across them. For a solution of solid solute in a liquid solvent, ideal SPM allow free movement of solvent particles across it, but not solute particles. These membranes contain a network of submicroscopic holes or pores through which small solvent molecules may pass but not the bigger solute particle.

7.4.3 Osmotic presssure (π) : It is the pressure which should be applied on the solution to just prevent osmosis or The hydrostatic pressure built up on the solution which just stops the osmosis.

osmotic pressure = hydrostatic pressure

$\pi = \rho gh$



The excess pressure equal to the osmotic pressure must be applied on the solution side to prevent osmosis

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7.4.4 Van't Hoff laws :

(i) The osmotic pressure (π) of a solution is directly proportional to its molar concentration(C), when the temperature is kept constant. (Van't Hoff-Boyle's law)

thus $\pi \propto C$ (when temperature is constant)

(ii) Concentration remaining same, the osmotic pressure of a dilute solution is directly proportional to its absolute temperature (T). (Van't Hoff-Charle's law)

$\pi \propto T$ (when C is constant)

Combining the two laws, i.e., when concentration and temperature both are changing, the osmotic pressure will be given by :

$\pi \propto C.T$ or $\pi = CRT$

Where R = Universal gas constant.

7.4.5 Isotonic or iso-osmotic solution :

Solutions which have the same osmotic pressures at a given temperature are called isotonic or iso-osmotic solutions

When isotonic solutions are separated by semipermeable membrane, no osmosis occurs between them. For example, the osmotic pressure associated with the fluid inside the blood cell is equivalent to that of 0.9% (mass/volume) sodium chloride solution, called normal saline solution and it is safe to inject intravenously. On the other hand, if we place the cells in a solution containing more than 0.9% (mass/volume) sodium chloride, water will flow out of the cells and they would shrink. Such a solution is called **hypertonic**. If the salt concentration is less than 0.9% (mass/volume), the solution is said to be **hypotonic**. In this case, water will flow into the cells if placed in this solution and they would swell.

Ex.18. ***A cane sugar solution has an osmotic pressure of 2.46 atm at 300 K. What is the molarity of the solution.***

Sol. $\pi = CRT \quad or \quad C = \frac{\pi}{RT} = \frac{2.46}{300 \times 0.0821} = 0.1\,M$

Ex.19 ***A solution containing 8.6 g urea in one litre was found to be isotonic with 0.5% (wt./vol.) solution of an organic, non-volatile solution. The molecular weight of unknown organic substance.***

Sol. *Solutions are isotonic*

so $\pi_1 = \pi_2$

$\frac{n_1}{V_1}RT = \frac{n_2}{V_2}RT$ *{R & T are constant}*

so, $\frac{n_1}{V_1} = \frac{n_2}{V_2}$

or $\left(\frac{w_1}{m_1 \times v_1}\right)_{urea} = \left(\frac{w_2}{m_2 \times v_2}\right)_{organic}$

or $\frac{8.6}{60 \times 1000} = \frac{0.5}{m_2 \times 100}$

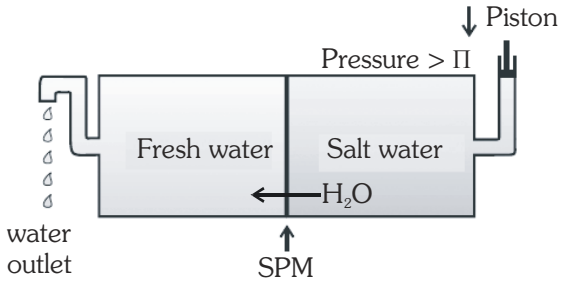
m_2 = *34.89 gm/mol*

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7.5 Reverse Osmosis and water purification :

If external pressure greater than osmotic pressure is applied, the flow of solvent molecules can be made to proceed from solution towards pure solvent, i.e., in reverse direction of the ordinary osmosis.

Reverse osmosis is used for the desalination of sea water for getting fresh drinking water.



Reverse osmosis occurs when a pressure larger than the osmotic pressure is applied to the solution.

8 Abnormal colligative properties :

It has been observed that difference in the observed and calculated molecular masses of solute is due to association or dissociation of solute molecules in solution. It results in a change in the number of particles in solution.

Van't Hoff in 1880, introduced a factor, called Van't Hoff factor (i). The factor 'i' is defined as

$$\mathbf{i} = \frac{\textbf{observed colligative property}}{\textbf{Calculated colligative property}} = \frac{\textbf{Normal molecular mass}}{\textbf{observed molecular mass}}.$$

$$= \frac{\text{total number of particles after dissociation / association}}{\text{Number of particles initially taken}}$$

In case of association of solute particles in solution, the observed molecular weight of solute being more than the normal, the value of factor 'i' is less than unity (i.e. $i < 1$), while for dissociation the value of i is greater than unity (i.e. $i > 1$), because the observed molecular weight has lesser value than normal molecular weight.

Values of van't Hoff factor, i, at Various Concentrations for NaCl, KCl, $MgSO_4$ and K_2SO_4

Salt	*Values of i			Van't Hoff factor i for complete dissociation of solute
	0.1 m	0.1 m	0.001 m	
NaCl	1.87	1.94	1.97	2.00
KCl	1.85	1.94	1.98	2.00
$MgSO_4$	1.21	1.53	1.82	2.00
K2SO_4	2.32	2.70	2.84	3.00

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8.1 Calculation of 'i' in case of dissociation

$$A_xB_y \longrightarrow x\,A^{y+} + y\,B^{x-}$$

Initial mol	1	0	0
after dissociation	$1-\alpha$	$x\alpha$	$y\alpha$

Total no. of solute particles $= 1-\alpha+x\alpha+y\alpha = 1-\alpha+\alpha(x+y)$

or $i = 1-\alpha+n\alpha$ [where $x+y=n$ (total ions.)]

or **$i = 1+\alpha(n-1)$**

For strong electrolytes : $\alpha = 1$ or 100%, so i = n (total no. of ions)

$NaCl \rightarrow Na^+ + Cl^-$ (i = 2)

$K_2SO_4 \rightarrow 2K^+ + SO_4^{2-}$ (i = 3)

For complex compound

$K_4[Fe(CN)_6] \rightarrow 4K^+ + [Fe(CN)_6]^{4-} \Rightarrow$ n = 5

S. No.	Solute type	Example	Ionisation	No. of particles in the solution from 1 mole solute (n)	van't Hoff factor ('i')	Abnormal molecular mass
1.	Non-electrolyte	Urea, sucrose, glucose, fructose	—	1	1	m_{normal}
2.	Binary electrolyte AB type	NaCl, KCl, HCl CH_3COOH NH_4OH, NaOH etc.	$AB \rightleftharpoons A^+ + B^-$ $1-\alpha \quad \alpha \quad \alpha$	2	$1+\alpha$	$\frac{m_{normal}}{1+\alpha}$
3.	Ternary electrolyte AB_2 type or A_2B type	$CaCl_2$, $BaCl_2$ H_2SO_4, $K_2[PtCl_6]$	$AB_2 \rightleftharpoons A^{2+} + 2B^-$ $1-\alpha \quad \alpha \quad 2\alpha$ $A_2B \rightleftharpoons 2A^+ + B^-$ $1-\alpha \quad 2\alpha \quad \alpha$	3 3	$1+2\alpha$ $1+2\alpha$	$\frac{m_{normal}}{1+2\alpha}$ $\frac{m_{normal}}{1+2\alpha}$
4.	Quanternary electrolyte AB_3 or A_3B type	$AlCl_3$, $K_3[Fe(CN)_6]$	$AB_3 \rightleftharpoons A^{3+} + 3B^-$ $1-\alpha \quad \alpha \quad 3\alpha$ $A_3B \rightleftharpoons 3A^+ + B^{3-}$ $1-\alpha \quad 3\alpha \quad \alpha$	4 4	$1+3\alpha$ $1+3\alpha$	$\frac{m_{normal}}{1+3\alpha}$ $\frac{m_{normal}}{1+3\alpha}$
5.	General electrolyte AB_{n-1}	One mole of solute giving 'n' ions in the solution	$AB_{n-1} \rightleftharpoons A^{+(n-1)} + (n-1)B^-$ $1-\alpha \quad \alpha \quad (n-1)\alpha$	n	$1+(n-1)\alpha$	$\frac{m_{normal}}{[1+(n-1)\alpha]}$

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8.2 Calculation of 'i' in case of assocation

	$nA \longrightarrow$	$(A)_n$
Initial mol	1	0
after dissociation	$1-\alpha$	α/n

$$i = \frac{1-\alpha+\frac{\alpha}{n}}{n}, \quad \alpha = \text{degree of assocation}$$

$$i = 1 + \alpha\left(\frac{1}{n}-1\right)$$

Ex.20. ***Determine the amount of $CaCl_2$ (i = 2.47) dissolved in 2.5 litre of water such that its osmotic pressure is 0.75 atm at 27°C.***

Sol. *We know that,*

$$\pi = i\frac{n}{V}RT \Rightarrow \pi = i\frac{w}{MV}RT \Rightarrow w = \frac{\pi MV}{iRT} = \frac{0.75\times111\times2.5}{2.47\times0.0821\times300} = 3.42\ g$$

Hence, the required amount of $CaCl_2$ is 3.42 g.

Ex.21 ***19.5 g of CH_2FCOOH is dissolved in 500 g of water. The depression in the freezing point of water observed is 1.0°C. Calculate the van't Hoff factor and dissociation constant of fluoroacetic acid.***

Sol. *It is given that :*

$w_1 = 500\ g$

$w_2 = 19.5\ g$

$K_f = 1.86\ K\ kg\ mol^{-1}$

$\Delta T_f = 1\ K$

We know that :

$$M_2 = \frac{K_f\times w_2\times1000}{\Delta T_f\times w_1} = \frac{1.86\,K\,kg\,mol^{-1}\times19.5\,g\times1000\,g\,kg^{-1}}{500\,g\times1K} = 72.54\ mol^{-1}$$

Therefore, observed molar mass of CH_2FCOOH, $(M_2)_{obs} = 72.54$ mol

The calculated molar mass of CH_2FCOOH is :

$(M_2)_{cal} = 14 + 19 + 12 + 16 + 16 + 1 = 78\ g\ mol^{-1}$

Therefore, van't Hoff factor, $i = \frac{(M_2)_{cal}}{(M_2)_{obs}} = \frac{78\,g\,mol^{-1}}{72.54\,g\,mol^{-1}} = 1.0753$

Let α be the degree of dissociation of CH_2FCOOH

	$CH_2FCOOH \rightleftharpoons$	CH_2FCOO^- +	H^+	
Initial conc.	*C mol L^{-1}*	*0*	*0*	
At equilibrium	*C (1 − α)*	*Cα*	*Cα*	*Total = C (1 + α)*

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$\therefore\ i = \frac{C(1+\alpha)}{C}$

$\Rightarrow i = 1 + \alpha \Rightarrow \alpha = i - 1$

$= 1.0753 - 1 = 0.0753$

Now, the value of K_a is given as :

$$K_a = \frac{[CH_2FCOO^-][H^-]}{[CH_2FCOOH]} = \frac{C\alpha.C\alpha}{C(1-\alpha)} = \frac{C\alpha^2}{1-\alpha}$$

Taking the volume of the solution as 500 mL. We have the concentration : 19.5 M

$C = \frac{78}{500} \times 1000\,M$ *= 0.5 M*

Therefore, $K_a = \frac{C\alpha^2}{1-\alpha}$

$= \frac{0.5 \times (0.0753)^2}{1-0.0753} = \frac{0.5 \times 0.00567}{0.9247}$ *= 0.00307 (approximately) =* 3.07×10^{-3}

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MISCELLANEOUS PREVIOUS YEAR QUESTIONS

1. To 500 cm^3 of water, 3×10^{-3} kg of acetic acid is added. If 23% of acetic acid is dissociated, what will be the depression in freezing point ? K_f and density of water are 1.86 K kg^{-1} mol^{-1} and 0.997 g cm^{-3} respectively. **[JEE 2000]**

Ans. 0.23 K

2. Match the boiling point with K_b for x, y and z, if molecular weight of x, y and z are same. **[JEE 2003]**

	b.pt.	**K_b**
x	100	0.68
y	27	0.53
z	253	0.98

Ans. $K_b(x) = 0.68$, $K_b(y) = 0.53$, $K_b(z) = 0.98$

3. During depression of freezing point in a solution, the following are in equilibrium **[JEE 2003]**

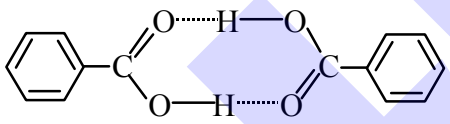
(A) liquid solvent-solid solvent (B) liquid solvent-solid solute
(C) liquid solute-solid solute (D) liquid solute-solid solvent

Ans. (A)

Sol. Equilibrium exists between liquid solvent & solid solvent.

4. 1.22 g of benzoic acid is dissolved in (i) 100 g acetone (K_b for acetone = 1.7) and (ii)100 g benzene (K_b for benzene = 2.6). The elevation in boiling points T_b is 0.17°C and 0.13°C respectively.
(a) What are the molecular weights of benzoic acid in both the solutions?
(b) What do you deduce out of it in terms of structure of benzoic acid? **[JEE 2004]**

Ans. (a)122, (b) It means that benzoic acid remains as it is in acetone while it dimerises in benzene

as 

Sol. (a)

(i) $\Delta T_b = i\, K_b \,.\, m$

$$0.17 = i \times 1.7 \times \frac{1.22/122}{0.1}$$

$$i = 1 = \frac{(\text{mol wt})_{Th}}{(\text{mol wt})_{ACt}}$$

$$(\text{mol wt})_{ACt} = \frac{122}{1} \text{ gm/mol}$$

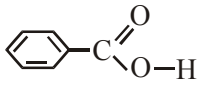
(ii) $\Delta T_b = i \times K_b \,.\, m$

$$0.13 = i \times 2.6 \times \frac{1.22/122}{0.1}$$

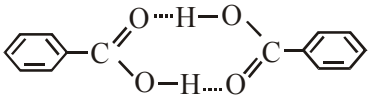
$$i = 0.5 = \frac{(\text{mol wt})_{Th}}{(\text{mol wt})_{ACt}} \Rightarrow (\text{mol ut})_{ACl} = \frac{122}{0.5}$$

$$= 244 \text{ gm/mol}$$

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(b) In Acetone its normal ie. 

but in benzene it is in dimenc form



5. A 0.004 M solution of Na_2SO_4 is isotonic with a 0.010 M solution of glucose at same temperature. The apparent degree of dissociation of Na_2SO_4 is

(A) 25% (B) 50% (C) 75% (D) 85% **[JEE 2004]**

Ans. (C)

Sol. $\pi_{Na_2So_4} = \pi_{g10C°}$

$i \times 0.004 = 0.010$

$i = 2.5 = 1 + 2\alpha \Rightarrow \alpha = 0.75$ or 75%

6. The elevation in boiling point, when 13.44 g of freshly prepared $CuCl_2$ are added to one kilogram of water, is [Some useful data, $K_b(H_2O) = 0.52$ kg K mol^{-1}, mol. wt. of $CuCl_2 = 134.4$ gm]

(A) 0.05 (B) 0.1 (C) 0.16 (D) 0.21 **[JEE 2005]**

Ans. (C)

Sol. $\Delta T_b = i\, K_b \,.\, m$

$= 3 \times 0.52 \times \dfrac{13.44/134.4}{1}$

$= 0.156$ K

$\simeq 0.16$ K

7. 72.5 g of phenol is dissolved in 1 kg of a solvent ($k_f = 14$) which leads to dimerization of phenol and freezing point is lowered by 7 kelvin. What percent of total phenol is present in dimeric form? **[JEE 2006]**

Ans. 35% (degree of asso = 70%)

Sol. $\underset{0.77\text{ mol}}{2\,PhOH} \rightleftharpoons (PhOH)_2$

$0.77(1-\alpha) \qquad 0.77\,\alpha/2$

$i = 1 - \alpha/2$

$\Delta T_f = i\, K_f \,.\, m$

$7 = (1-\alpha) \times 14 \times \dfrac{72.5/94}{1}$

$0.649 = 1 - \alpha/2$

$\alpha/2 = 0.35$

$\alpha = 0.70$

% of phenol in dimeric form

$= \dfrac{0.77\,\alpha/2}{0.77} \times 100 = \dfrac{\alpha}{2} \times 100 = 35\%$

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8. When 20 g of naphtholic acid ($C_{11}H_8O_2$) is dissolved in 50 g of benzene (K_f = 1.72 K kg mol^{-1}), a freezing point depression of 2 K is observed. The van't Hoff factor (*i*) is **[JEE 2007]**

(A) 0.5 (B) 1 (C) 2 (D) 3

Ans. (A)

Sol. $\Delta T_f = i\, K_f \cdot \dfrac{n}{W/1000}$

$2 = i \times 1.72 \times \dfrac{20/172}{50/1000}$

$i = 0.5$

Paragraph for Question No. Q.9 to Q.11

Properties such as boiling point, freezing point and vapour pressure of a pure solvent change when solute molecules are added to get homogeneous solution. These are called colligative properties. Applications of colligative properties are very useful in day-to-day life. One of its examples is the use of ethylene glycol and water mixture as anti-freezing liquid in the radiator of automobiles.

A solution **M** is prepared by mixing ethanol and water. The mole fraction of ethanol in the mixture is 0.9.

Given:

Freezing point depression constant of water $\left(K_f^{water}\right) = 1.86$ K kg mol^{-1}

Freezing point depression constant of ethanol $\left(K_f^{ethanol}\right) = 2.0$ K kg mol^{-1}

Boiling point elevation constant of water $\left(K_b^{water}\right) = 0.52$ K kg mol^{-1}

Boiling point elevation constant of ethanol $\left(K_b^{ethanol}\right) = 1.2$ K kg mol^{-1}

Standard freezing point of water = 273 K

Standard freezing point of ethanol = 155.7 K

Standard boiling point of water = 373 K

Standard boiling point of ethanol = 351.5 K

Vapour pressure of pure water = 32.8 mm Hg

Vapour pressure of pure ethanol = 40 mm Hg

Molecular weight of water = 18 g mol^{-1}

Molecular weight of ethanol = 46 g mol^{-1}

In answering the following questions, consider the solutions to be ideal dilute solutions and solutes to be non-volatile and non-dissociative.

9. The freezing point of the solution **M** is **[JEE 2008]**

(A) 268.7 K (B) 268.5 K (C) 234.2 K (D) 150.9 K

Ans. (D)

$(T_f')_{Alc.sol^n} = (T_f)_{EtoH} - (K_f)_{etoh} .m$

$= 115.7 - 2 \times \dfrac{0.1}{0.9 \times 46/1000}$

$= 155.7 - 2 \times 2.415$

$= 150.9$ K

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10. The vapour pressure of the solution **M** is **[JEE 2008]**

(A) 39.3 mm Hg (B) 36.0 mm Hg (C) 29.5 mm Hg (D) 28.8 mm Hg

Ans. (B)

Sol. $P^s = (P^\circ)_{solvent} \cdot X_{solvent}$

$= 40 \times 0.9$

$= 36.0$ mm Hg

11. Water is added to the solution **M** such that the mole fraction of water in the solution becomes 0.9. The boiling point of this solution is **[JEE 2008]**

(A) 380.4 K (B) 376.2 K (C) 375.5 K (D) 354.7 K

Ans. (B)

Sol. $(T_b')_{Aq\,sol^n} = (T_b)_{solvent\,H_2O} + K_b \cdot m$

$= 373 + 0.52 \times \dfrac{0.1}{0.9 \times 18/1000}$

$= 373 + 0.52 \times 6.173$

$= 376.2$ K

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